

Linear differential equations with constant coefficients

Linear differential equations with constant coefficients

The general linear ordinary differential equation of order n is

$$P_0 \frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_{n-1} \frac{dy}{dx} + P_n y = f(x) \quad \rightarrow (1)$$

$$P_0 \neq 0$$

$$P_1, P_2, \dots, P_n \quad \text{real functions or constants}$$

$$f(x) - \quad \text{is a function of } x \text{ only.}$$

$$f(x) \neq 0 \quad \text{nonhomogeneous}$$

Examples

$$\frac{d^3 y}{dx^3} + 2 \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} - y = \sin(x) \quad \leftarrow \text{nonhomogeneous}$$

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2 y = 0 \quad \leftarrow \text{homogeneous}$$

$$P_0 \frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_{n-1} \frac{dy}{dx} + P_n y = f(x) \quad \rightarrow (1)$$

A differential equation has constant coefficients if only constant functions appear as coefficients in the equation (1).

D - Differential operator

$$D = \frac{d}{dx} \quad \text{- the differential operator}$$

$$Dy = \frac{dy}{dx} \quad \text{Differentiation with respect to } x$$

$$D^2 y = \frac{d^2 y}{dx^2}$$

D^2 allows to obtain the second derivative of the function y .

Superscript gives what the derivative is.

$$D^3 y = \frac{d^3 y}{dx^3} \dots \quad D^n y = \frac{d^n y}{dx^n}$$

$$P_0 \frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_{n-1} \frac{dy}{dx} + P_n y = f(x) \quad \rightarrow (1)$$

Using the differential operator D , we can write (1) in the form

$$P_0 D^n y + P_1 D^{n-1} y + \dots + P_{n-1} D y + P_n y = f(x)$$

$$(P_0 D^n + P_1 D^{n-1} + \dots + P_{n-1} D + P_n) y = f(x)$$



$$F(D)$$

$$F(D)y = f(x)$$

Here $F(D)$ is a polynomial differential operator in D of order n .

Operator rules

If $\phi(D)$ and $\varphi(D)$ are polynomial operators

Sum rule $[\phi(D) + \varphi(D)]y(x) = \phi(D)y(x) + \varphi(D)y(x)$

Linearity rule

$$\phi(c_1 y_1(x) + c_2 y_2(x)) = c_1 (\phi y_1(x)) + c_2 \phi y_2(x)$$

c_1 and c_2 are constants.

Index rule

$$D^m D^n = D^n D^m = D^{m+n}$$

Also $D(D + 1) = D^2 + D$

Take $y = y(x)$

$$\phi D(y) = D(D + 1)y$$

$$\frac{d}{dx} \left(\frac{d}{dx} + 1 \right) y = \frac{d^2 y}{dx^2} + \frac{dy}{dx}$$

$$\begin{aligned} \phi D(y) &= (D^2 + D)y \\ &= \left(\frac{d^2}{dx^2} + \frac{d}{dx} \right) y \\ &= \frac{d^2 y}{dx^2} + \frac{dy}{dx} \end{aligned}$$

$$\phi(D)y = \phi(D)y$$

Substitution rule

Differential operator to exponentials

$$F(D) e^{ax} = F(a) e^{ax}$$

$$D e^{ax} = \frac{d}{dx} e^{ax} = a e^{ax} \quad D^2 e^{ax} = a^2 e^{ax} \quad D^3 e^{ax} = a^3 e^{ax} \dots D^r e^{ax} = a^r e^{ax}$$

$$F(D) = P_0 D^n + P_1 D^{n-1} + \dots + P_n$$

$$\begin{aligned} F(D) e^{ax} &= \{P_0 D^n + P_1 D^{n-1} + \dots + P_n\} e^{ax} \\ &= P_0 D^n e^{ax} + P_1 D^{n-1} e^{ax} + \dots + P_n e^{ax} \\ &= P_0 a^n e^{ax} + P_1 a^{n-1} e^{ax} + \dots + P_n e^{ax} \\ &= (P_0 a^n + P_1 a^{n-1} + \dots + P_n) e^{ax} \end{aligned}$$

$$\underline{\underline{F(D) e^{ax} = F(a) e^{ax}}}$$

$$(D^2 + 1) e^{2x} = (2^2 + 1) e^{2x} = 5 e^{2x}$$

Exponential –shift rule

$$F(D) e^{ax} v(x) = e^{ax} F(D + a) v(x) \rightarrow (1)$$

If $F(D) = D = \frac{d}{dx}$

Using product rule
for derivatives

$$D e^{ax} v(x) = e^{ax} Dv(x) + a e^{ax} v(x) = e^{ax} (D + a) v(x)$$

Apply (1) to D repeatedly

$$F(D) = D^2$$

$$\begin{aligned} D^2 e^{ax} v &= D(D e^{ax} v) = D(e^{ax} (D + a) v) = e^{ax} D(D + a) v + (D + a) v e^{ax} \\ &= e^{ax} (D + a)^2 v \end{aligned}$$

In the same way $F(D) = D^3$

$$D^3 e^{ax} v(x) = e^{ax} (D + a)^3 v(x)$$

Exponential –shift rule...

$$F(D) = P_0 D^n + P_1 D^{n-1} + \dots + P_n \rightarrow (2)$$

Now we show equation (1) is true the operator form $D^k e^{ax} v(x)$

You can observe that coefficient is a^k . Multiply the Equation (2) by a^k and sum these equations for the different values of $k = 1 \dots n$.

$$F(D) e^{ax} v(x) = e^{ax} F(D + a) v(x)$$

Example

$$\begin{aligned} D^2 e^{2x} \sin x &= e^{2x} (D + 2)^2 \sin x \\ &= e^{2x} (D^2 + 4D + 4) \sin x \\ &= e^{2x} (-\sin x + 4 \cos x + 4 \sin x) \\ &= e^{2x} (3 \sin x + 4 \cos x) \end{aligned}$$

Higher order homogeneous linear ODE's with constant coefficients.

$$F(D)y = 0 \rightarrow (1)$$

If $y = e^{mx}$ is a solution to equation (1), for some real or complex m .

Using substitution rule

$$F(D)e^{mx} = F(m)e^{mx} = 0$$

$$e^{mx} \neq 0 \Rightarrow F(m) = 0$$

$$P_0 m^n + P_1 m^{n-1} + \dots + P_n = 0$$

$F(m)$ is the auxiliary polynomial

$F(m) = 0$ characteristic equation or auxiliary equation and roots are called characteristic roots.

$$F(D)e^{mx} = F(m)e^{mx} = 0$$

$$e^{mx} \neq 0 \Rightarrow F(m) = 0$$

$$P_0 m^n + P_1 m^{n-1} + \dots + P_n = 0$$

We can write down the solution of the equation $F(D)y = 0$ in the form $y = Ae^{mx}$.

$$F(D)e^{mx} = 0$$

$$F(m)e^{mx} = 0 \quad (\text{Theorem 1})$$

$$e^{mx} \neq 0 \quad \forall x \in \mathbb{R}$$

$$F(m) = 0$$

$$P_0 m^n + P_1 m^{n-1} + P_2 m^{n-2} + \dots + P_n = 0$$

This equation is called the **Auxiliary Equation**. The roots of it are called **Characteristic Roots**.

Note :-

(1) If the characteristic Roots $m_1, m_2, \dots, m_n$ are distinct then the complementary function is

$$y = A_1 e^{m_1 x} + A_2 e^{m_2 x} + \dots + A_n e^{m_n x}$$

(2) If r of $m_1, m_2, \dots, m_n$ are equal then the complementary function is

$$y = A_1 e^{mx} + A_2 x e^{mx} + A_3 x^2 e^{mx} + \dots + A_r x^{r-1} e^{mx} + A_{r+1} e^{m_{r+1}x} + \dots + A_n e^{m_n x}$$

Complementary function

A function of x which satisfies the linear differential equation $F(D)y = 0$ is known as the complementary function of the linear equation.

Roots are Real and Distinct $m_1, m_2, \dots, m_n$

If there are **n distinct roots** to the characteristic equation

$$y = A_1 e^{m_1 x} + A_2 e^{m_2 x} + \dots + A_n e^{m_n x}$$

$$m_1 = 1, \quad m_2 = 2 \quad C.F \quad y = A_1 e^x + A_2 e^{2x}$$

Equal Roots

If r roots are equal and others are distinct

$$\underbrace{m_1 = m_2 = \dots = m_r}_{= m} \neq m_{r+1} \neq \dots \neq m_n$$

then Complementary function

$$y = A_1 e^{mx} + A_2 x e^{mx} + A_3 x^2 e^{mx} + \dots + A_r x^{r-1} e^{mx} + A_{r+1} e^{m_{r+1}x} + \dots + A_n e^{m_n x}$$

Theorem

The differential equation $(D - m)^r y = 0$ has independent solutions

$$e^{mx}, xe^{mx}, x^2 e^{mx}, \dots, x^{r-1} e^{mx}$$

$$m_1 = 1, \quad m_2 = 1 \quad C.F \quad y = A_1 e^x + A_2 x e^x$$

$$m_1 = 0, \quad m_2 = 0 \quad C.F \quad y = A_1 e^{0 \cdot x} + A_2 x e^{0 \cdot x} = A_1 + A_2 x$$

$$m_1 = 2, \quad m_2 = 2 \quad m_3 = 2, \quad m_4 = -1$$

$$C.F \quad y = A_1 e^{2x} + A_2 x e^{2x} + A_3 x^2 e^{2x} + A_4 e^{-x}$$

Complex-Conjugate Roots

If roots are complex conjugate $m_1 = p + iq$, $m_2 = p - iq$

$$y = A_1 e^{(p+iq)x} + A_2 e^{(p-iq)x}$$

$$y = e^{px} [A_1 e^{iqx} + A_2 e^{-iqx}]$$

$$e^{ix} = \cos x + i \sin x$$

$$y = e^{px} [A_1 (\cos qx + i \sin qx) + A_2 (\cos qx - i \sin qx)]$$

Complementary function

$$y = e^{px} (B_1 \cos qx + B_2 \sin qx)$$

$$B_1 = A_1 + A_2$$

$$B_2 = i(A_1 - A_2)$$

Example 1

$$(D^2 - 2D + 1)y = 0$$

Let $y = e^{mx}$ be a solution.

The characteristic equation is

$$m^2 - 2m + 1 = 0$$

$$\Rightarrow (m - 1)^2 = 0$$

$$\Rightarrow m = 1 \text{ (twice)}$$

Complementary function

$$\begin{aligned} y &= Ae^x + Bxe^x \\ &= e^x(A + Bx) \end{aligned}$$

Example 2 $(D^3 - 6D^2 + 11D - 6)y = 0$

Let $y = e^{mx}$ be a solution.

$$(D^3 - 6D^2 + 11D - 6)e^{mx} = 0$$

The characteristic equation is

$$(m^3 - 6m^2 + 11m - 6)e^{mx} = 0$$

$$e^{mx} \neq 0 \quad \text{for } \forall x \in \mathbb{R}$$

$$m^3 - 6m^2 + 11m - 6 = 0$$

$$(m - 1)(m - 2)(m - 3) = 0$$

characteristic roots $m = 1, 2, 3$

Complementary function $y = A_1 e^x + A_2 e^{2x} + A_3 e^{3x}$

A_1, A_2, A_3 are arbitrary constants. 19

Example 3

$$(D^3 - 1)e^{-3x} = [(-3)^3 - 1]e^{-3x}$$

$$(D^3 + 1)\sinh x = (D^3 + 1)\left(\frac{e^x - e^{-x}}{2}\right)$$

$$= \frac{1}{2}(D^3 + 1)e^x - \frac{1}{2}(D^3 + 1)e^{-x}$$

$$= \frac{1}{2}(1^3 + 1)e^x - \frac{1}{2}((-1)^3 + 1)e^{-x}$$

$$= \frac{1}{2}(1^3 + 1)e^x$$

$$= e^x$$

Example 4

Solve $(D^4 - 1)y = 0$

$$y = e^{m x}$$

$$(D^4 - 1)e^{m x} = 0$$

The characteristic equation is

$$(m^4 - 1)e^{m x} = 0$$

$$(m^2 - 1)(m^2 + 1) = 0$$

$$m = \pm 1, m = \pm i$$

Complementary function

$$y = A_1 e^{-i x} + A_2 e^{i x} + A_3 e^x + A_4 e^{-x}$$

$$= A_1 [\cos x + i \sin x] + A_2 [\cos x - i \sin x] + A_3 e^x + A_4 e^{-x}$$

Example 5

$$(D^2 - 2D + 1)y = 0 ; y = e^{mx}$$

The characteristic equation is

$$(m^2 - 2m + 1)y = 0$$

$$m = -1, -1$$

Complementary function

$$y = A_1 e^{-x} + A_2 x e^{-x}$$

Example 6

$$(D + 1)^3 (D - 4)y = 0$$

$$y = e^{mx}$$

$$(m + 1)^3 (m - 4) e^{mx} = 0$$

$$(m + 1)^3 (m - 4) = 0$$

Differential Operator to Trigonometric Functions

$$F(D^2) \cos ax = F(-a^2) \cos ax$$

$$D(\cos ax) = \frac{d}{dx} \cos ax = -a \sin ax$$

$$D^2 \cos ax = -a^2 \cos ax$$

$$D^3 \cos ax = a^3 \sin ax$$

$$D^4 \cos ax = a^4 \cos ax$$

$$F(D^2) = P_0 D^{2n} + P_1 D^{2(n-1)} + \dots + P_{n-1} D^2 + P_n$$

$$F(D^2) \cos ax = F(-a^2) \cos ax$$

$$\begin{aligned}
F(D^2) \cos ax &= P_0 D^{2n} \cos ax + P_1 D^{2(n-1)} \cos ax + \dots + P_{n-1} D^2 \cos ax + P_n \cos ax \\
&= \cos ax \left[P_0 (-a^2)^n + P_1 (-a^2)^{n-1} + \dots + P_n \right] \\
&= \cos ax F(-a^2)
\end{aligned}$$

$$F(D^2) \sin ax = F(-a^2) \sin ax$$

$$\begin{aligned}
F(D^2) \sin ax &= P_0 D^{2n} \sin ax + P_1 D^{2(n-1)} \sin ax + \dots + P_{n-1} D^2 \sin ax + P_n \sin ax \\
&= \sin ax \left[P_0 (-a^2)^n + P_1 (-a^2)^{n-1} + \dots + P_n \right] \\
&= \sin ax F(-a^2)
\end{aligned}$$